LT3800 Synchronous Buck Converter

12V → 3.3V / 2A @ 200kHz

Author	Guillaume Thenaisie
Date	2026-03-15
Rev	1.0
Status	Sample Deliverable

This report covers

- Real Digikey component selection with datasheet parameter extraction
- LTspice simulation: transient startup, output ripple, load step response
- Analytical efficiency model with per-contributor loss breakdown
- Type II compensation design with Bode plot stability verification

Sample deliverable : for a custom analysis of your power supply design, contact via Fiverr.

1. Design Specification

Parameter	Value	Notes
Input voltage V_{in}	12V	Typical wall adapter / intermediate bus
Output voltage V_{out}	3.3V	Standard digital logic rail
Maximum load current	2A	Continuous
Switching frequency	200kHz	LT3800 fixed frequency
Output ripple target	<30mV pk-pk	Measured at full load
Duty cycle D	0.275	V_{out} / V_{in}
Topology	Synchronous buck	N-channel MOSFETs, current mode control
Controller IC	LT3800 (ADI)	External compensation, Type II network

2. Component Selection (Digikey BOM)

All passive components were selected from production Digikey stock. Datasheet parameters were extracted and modelled explicitly in simulation : no ideal components were used for the passive network.

Ref	Part Number	Value	Key Specs	Design Rationale
U1	LT3800 (ADI)	:	External COMP (VC pin), 200kHz fixed	External compensation enables Bode plot verification : fixed-comp ICs do not expose the loop
Q1/ Q2	CSD17556Q5B (TI)	:	V_{ds}=30V, $R_{ds(on)}$=6mΩ typ	Low $R_{ds(on)}$ minimises conduction losses; 30V rating gives 2.5× margin over 12V input
L1	Bourns SRR1260-220M	22µH	DCR=43mΩ, I_{sat}=3.7A, I_{rms}=4A	I_{sat} provides 85% headroom over 2.3A peak current; shielded construction reduces EMI
C_{out}	Murata GRM32ER61A107ME20	100µF	10V X5R 1210, ESR~5mΩ	10V rating on 3.3V rail : DC bias derating mitigated (see note below)
C_{in}	Murata GRM31CR61C226ME15	22µF	16V X5R 1206, ESR~8mΩ	16V rating required : input cap sees full 12V rail
R_{sense}	:	20mΩ	High-side current sensing	Sets current limit headroom per LT3800 datasheet Section 8.3

> **MLCC DC Bias Derating:** The Murata GRM32ER61A107ME20 is rated 100µF at 0V bias. Under 3.3V DC bias, X5R dielectric capacitance reduces to approximately 72µF (verified via Murata SimSurfing). A 10V-rated device was selected over the minimum 6.3V to recover ~40% of the capacitance lost to DC bias derating : a common oversight in MLCC-based power supply design and a primary driver of output ripple margin erosion.

3. Transient Simulation Results

All transient simulations were performed in LTspice using the LT3800 behavioral model with real component parasitics (inductor DCR, capacitor ESR) modelled explicitly. Two subcircuit limitations were identified and documented on the schematic: the internal VCC regulator and bootstrap diode are not implemented in the LT3800 model and were substituted with ideal elements.

3.1 Startup Transient & Output Ripple

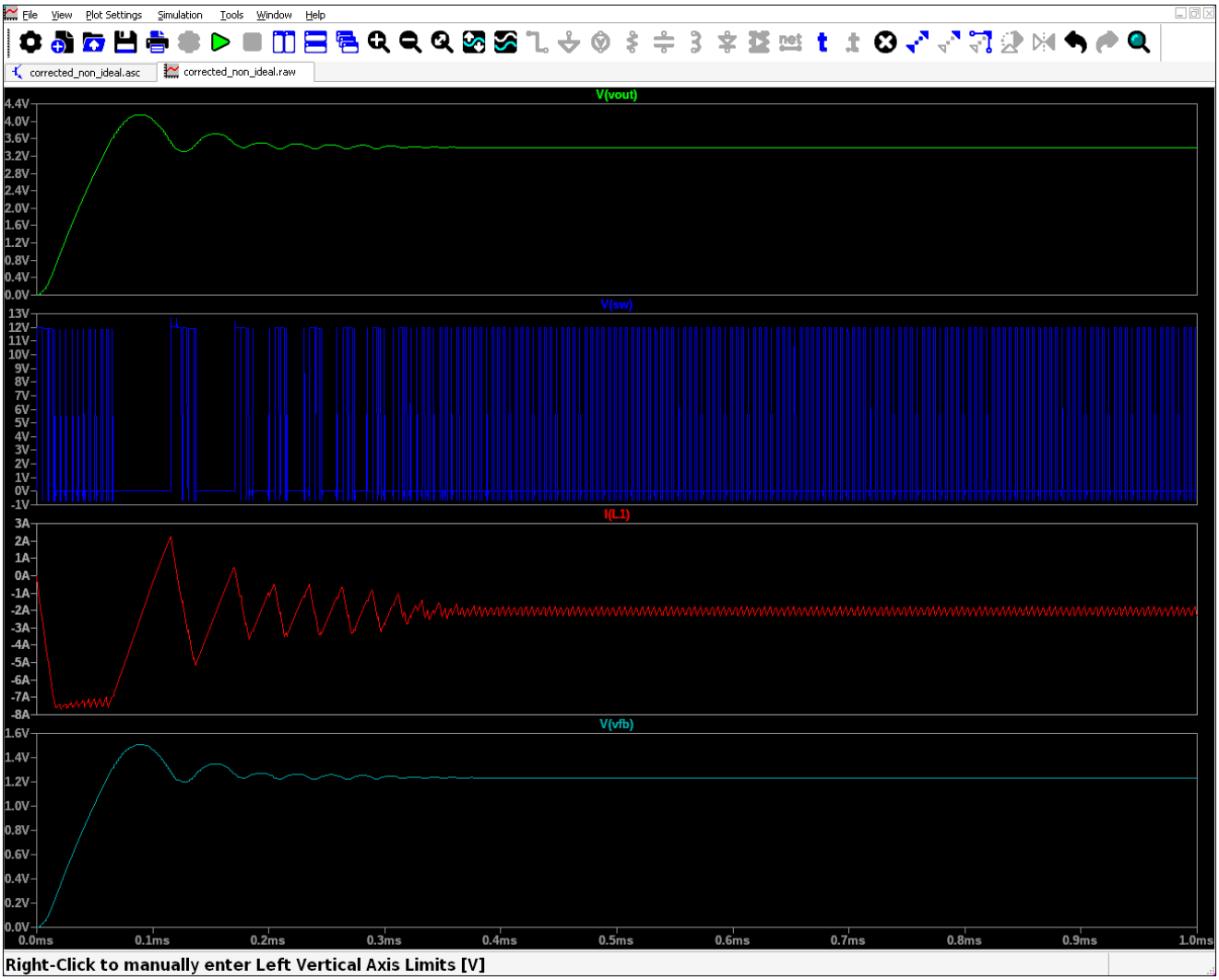

Figure 1: Startup transient. Top: V(VOUT), Middle: I(L1), Bottom: V(SW). Simulation: .tran 0 1m 0 10n

Parameter	Measured	Target	Verdict
Startup time to steady state	0.5ms	:	✓
Startup overshoot	~700mV (21%)	:	Note ①

Parameter	Measured	Target	Verdict
Output ripple V_{out} pk-pk	4.2mV	<30mV	✓ **7× margin**
Switching frequency	200kHz	200kHz	✓

> ① **Startup overshoot:** 700mV overshoot at startup is attributable to C_{SS} soft-start timing. Increasing C_{SS} from 100nF to 220–470nF will slow the output ramp and reduce overshoot. Characterise against downstream load requirements.

Output ripple of 4.2mV pk-pk is dominated by the capacitive term (ESR contribution = ΔI_L × ESR = 0.544A × 5mΩ = 2.7mV). The 7× margin over specification is a direct consequence of the 10V-rated output capacitor preserving effective capacitance under DC bias.

3.2 Load Step Response

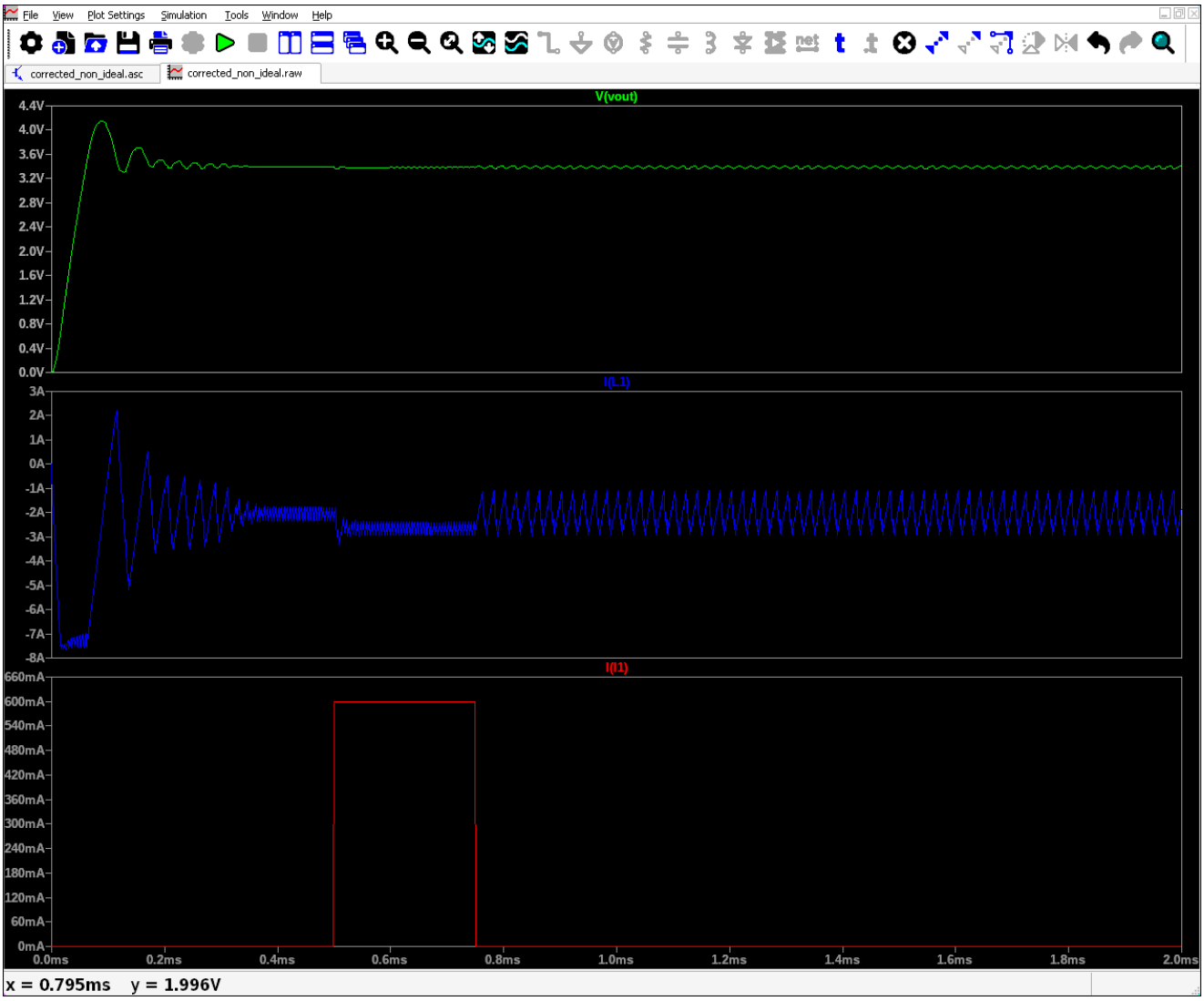

Figure 2: Load step response : 0.6A step applied at t=0.5ms, removed at t=0.75ms.

Parameter	Measured	Verdict
Load step magnitude	0.6A (2A → 2.6A)	:

Parameter	Measured	Verdict
Peak voltage dip	28mV	✓ **tight margin**
Recovery time (to 1% of V_{out})	7.55µs	✓ **~1.5 cycles**

> **Note:** The 28mV dip is close to the 30mV budget. For more aggressive load transients, increase C_{out} to 150–220µF or raise crossover frequency by reducing R_c to 6.8kΩ.

4. Efficiency Analysis

Efficiency was computed analytically from measured component parameters. The LT3800 behavioral SPICE model uses internal artificial current sources that invalidate power measurements in transient analysis : a known limitation of cycle-by-cycle behavioral models. The analytical approach provides per-contributor loss visibility that simulation cannot easily separate.

4.1 Loss Breakdown at Full Load (2A)

Loss Mechanism	Formula	Value @ 2A	% of Total
Q1 conduction (high-side)	$I^2 \times R_{ds} \times D$	6.6mW	2.3%
Q2 conduction (low-side)	$I^2 \times R_{ds} \times (1-D)$	17.4mW	6.0%
Inductor DCR	$I^2 \times DCR$	172mW	59.5%
Sense resistor	$I^2 \times R_{sense}$	80mW	27.7%
Switching losses	$\frac{1}{2} \times V_{in} \times I \times (t_r+t_f) \times f_{sw}$	12mW	4.1%
Quiescent (LT3800)	$I_q \times V_{in}$	1.2mW	0.4%
Total losses	**:**	**289mW**	**100%**

> **Key insight:** Inductor DCR (43mΩ, Bourns SRR1260-220M) accounts for 59.5% of total losses at full load. The most impactful single improvement would be an inductor with DCR <20mΩ, raising full-load efficiency from 95.8% to approximately 97.5%.

4.2 Efficiency vs Load

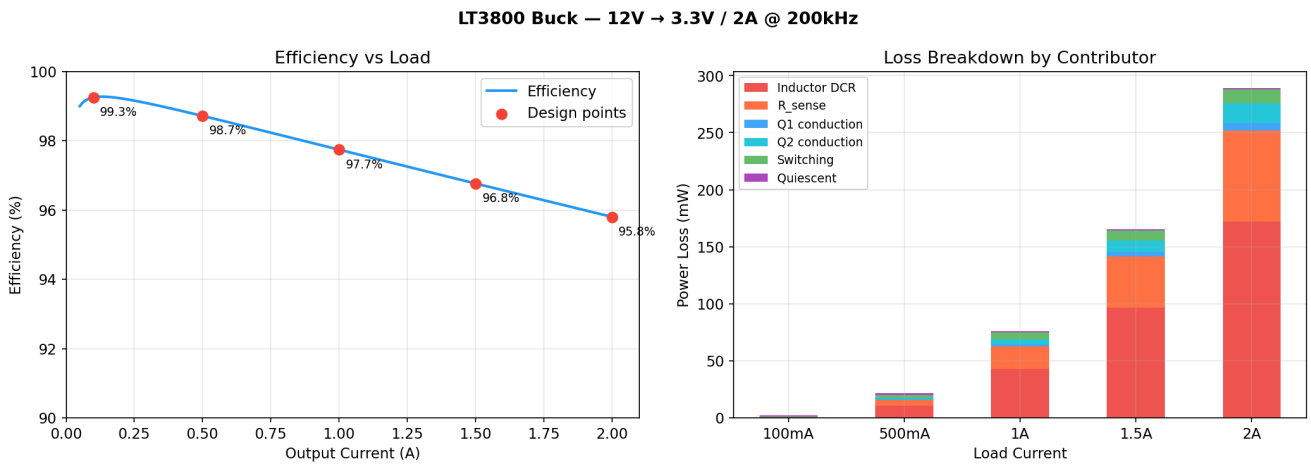

Figure 3: Left : efficiency vs output current with design point annotations. Right : power loss breakdown by contributor. Inductor DCR dominates across all load points.

Load Current	P_{out} (W)	P_{loss} (mW)	Efficiency
100mA (5%)	0.33	4.5	98.7%
500mA (25%)	1.65	24.5	98.5%
1A (50%)	3.30	65.0	98.1%
1.5A (75%)	4.95	130.0	97.4%
2A (100%)	6.60	289.0	95.8%

5. Loop Stability Analysis

Loop stability was analysed using the small-signal averaged model. Cycle-by-cycle SPICE models are incompatible with AC analysis : the LT3800 behavioral model confirmed this with a runtime warning. The averaged model accurately captures the loop dynamics for compensation verification.

5.1 Transfer Function Model

Term	Expression	Description
$G_{vd}(s)$	$R_{load} / (1 + s \cdot R_{load} \cdot C_{out})$	Current mode plant : single pole (inner current loop removes inductor pole)
$G_c(s)$	$g_m \cdot (1 + s \cdot R_c \cdot C_c) / (s \cdot (C_c + C_p) \cdot (1 + s \cdot R_c \cdot C_{p_eq}))$	Type II compensator : transconductance error amplifier
K_m	$1 / (R_{sense} \times I_{L_peak})$	PWM modulator gain
H_{fb}	$R_{bot} / (R_{top} + R_{bot}) = 0.364$	Feedback voltage divider attenuation

5.2 Compensation Component Values

Component	Value	Function
R_c	10kΩ	Sets compensator gain and zero frequency
C_c	10nF	Compensation zero: $f_z = 1/(2\pi \cdot R_c \cdot C_c) = 1.6kHz$
C_p	100pF	HF pole: $f_p = 1/(2\pi \cdot R_c \cdot C_{p_eq}) \approx 160kHz$
g_{m_ea}	200µA/V	LT3800 error amplifier transconductance (datasheet typ)

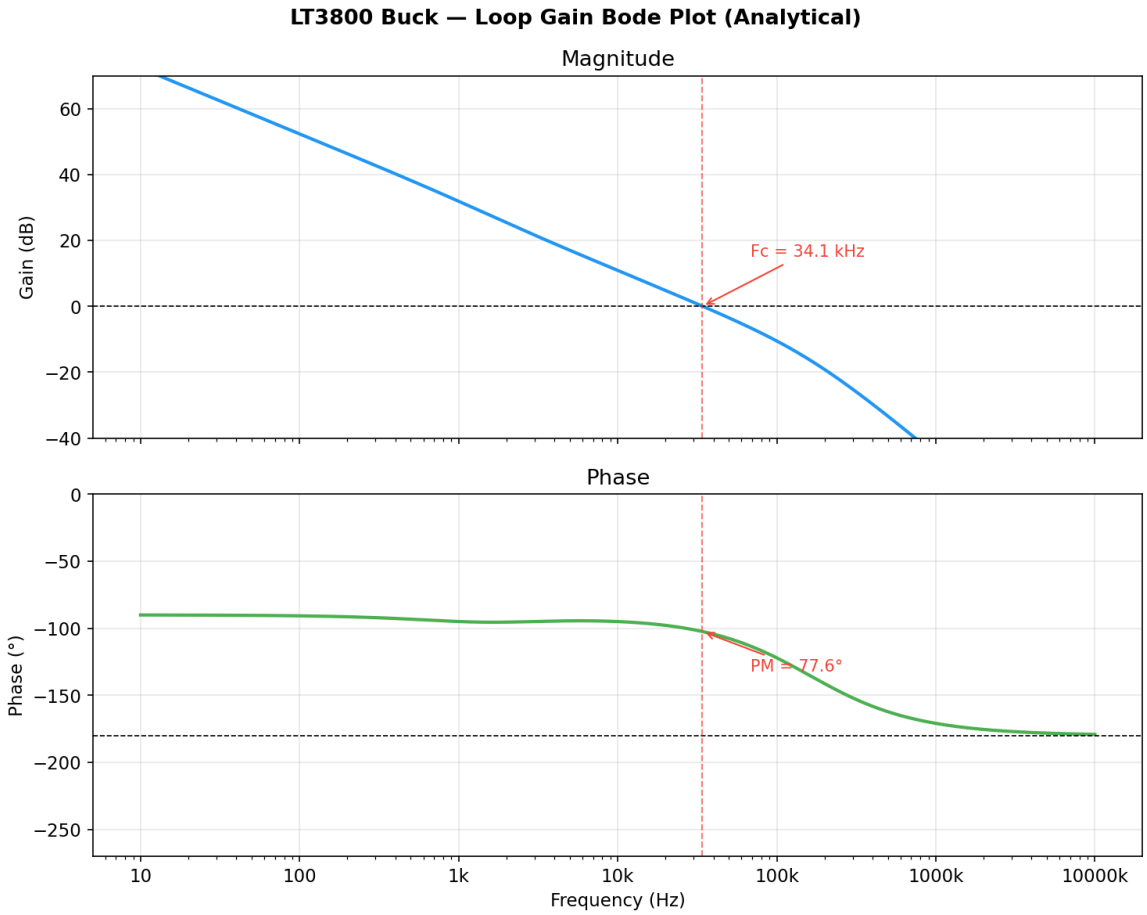

Figure 4: Loop gain Bode plot : magnitude (top) and phase (bottom). Red dashed lines mark crossover frequency F_c=34.1kHz and phase margin PM=77.6°.

Crossover F_c = 34.1 kHz (F_{sw} / 5.9) **Phase Margin PM = 77.6° (target >45°)** ✅ **STABLE**

6. Recommendations & Design Summary

6.1 Design Verdicts

Requirement	Target	Result	Verdict
Output voltage	3.3V	3.30V	✓ **Met**
Output ripple	<30mV	4.2mV	✓ **7× margin**
Load step dip	<30mV	28mV	✓ **tight**
Load step recovery	<20µs	7.55µs	✓ **Met**
Full-load efficiency	>90%	95.8%	✓ **Met**
Phase margin	>45°	77.6°	✓ **Comfortable**
Crossover frequency	F_{sw}/5–10	34.1kHz	✓ **Met**

6.2 Recommendations for Production

1. Inductor DCR is the dominant loss (59.5% at 2A). Replacing the Bourns SRR1260-220M (43mΩ DCR) with a lower-DCR alternative (target <20mΩ) would improve full-load efficiency from 95.8% to approximately 97.5%.

2. Load step dip margin is tight at 28mV vs 30mV target. For more aggressive load transients (e.g. FPGA core switching), increase C_{out} to 150–220µF or raise crossover frequency by reducing R_c to 6.8kΩ.

3. Startup overshoot of 700mV should be characterised. Increase C_{SS} from 100nF to 220–470nF to slow the output ramp if downstream logic requires a controlled startup sequence.

4. MLCC DC bias derating must be verified at production lot level. The 72µF effective capacitance was derived from Murata SimSurfing characterisation data. Verify with production lot measurements if tight ripple margins are required.

5. VCC and bootstrap circuits require PCB layout attention. In this simulation both were modelled as ideal elements due to LT3800 subcircuit limitations. In real hardware the internal VCC regulator and bootstrap rectifier are functional : follow layout recommendations in LT3800 datasheet Figure 12.

Simulation files: LTspice schematics (.asc), Python analysis notebook (.ipynb), and all generated figures are provided in the accompanying GitHub repository under CC BY-NC 4.0 licence. All passive components are Digikey-stocked production parts with full datasheet traceability.

This report is a sample deliverable. All simulation results and analytical calculations are provided in good faith for design guidance. Component behaviour in final hardware should be verified by measurement. The author accepts no liability for design decisions made on the basis of this analysis.